NATIONAL INSTITUTE OF TECHNOLOGY PATNA

Road Damage Detection and Classification Using Deep Neural Networks

Project implementation and preliminary local evaluation

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Presentation Outline

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Motivation behind the Work

Support road inspection with image-based damage detection.

- Manual road surveys require time and consistent human judgement.
- Cracks and potholes vary in shape, size, lighting and surrounding texture.
- A detector can locate and classify multiple damaged regions in one image.
- A lightweight model and simple interface make the prototype easier to test on ordinary computing hardware.

Source: Zeng & Zhong (2024), introduction and motivation [1].

Introduction: Four Target Damage Classes



D00
Longitudinal crack



D10
Transverse crack



D20
Alligator crack



D40
Pothole

Task: predict a bounding box, damage class and confidence for each detected region.

Source: RDD2022 images in local test samples [2]; examples show dataset content, not model predictions.

Literature Review

Work	Contribution	Relevance to this project
RDD2022 [2]	Road damage images from six countries with four target damage categories.	Common annotation scheme and realistic road scenes.
YOLOv8n [3]	Compact, single-stage object detector with multi-scale predictions.	Pretrained baseline for the local prototype.
Zeng & Zhong, 2024 [1]	YOLOv8-PD combines attention, Ghost feature extraction and a shared detection head.	Main reference for the proposed model variants.

Source: Selected paper [1]; RDD2022 benchmark [2]; pinned Ultralytics implementation [3].

Literature Review: What YOLOv8-PD Changes

Component	Purpose in the paper	Intended benefit
BOT	Add global context through attention.	Relate separated or elongated crack features.
LSKA	Use large separable kernels for spatial attention.	Capture a broader region of road texture.
C2fGhost	Generate features with cheaper operations.	Reduce feature-extraction cost in the neck.
LSCD	Share convolutional detection layers across scales.	Reduce detection-head parameters.

The paper balances detection performance against model size and computation.

Source: Zeng & Zhong (2024), proposed methodology [1].

Identified Gaps

- **Small and multiple potholes remain difficult.** The reference paper reports missed detections and lower AP for D40.
- **Class imbalance limits evaluation.** The local validation set has only nine D10 objects.
- **Reproduction needs care.** Some attention and head details are insufficiently specified for an exact replica.
- **Comparable evidence is missing locally.** A short baseline run cannot establish the value of individual modifications.

Source: Paper discussion [1]; local dataset audit; project architecture notes [4].

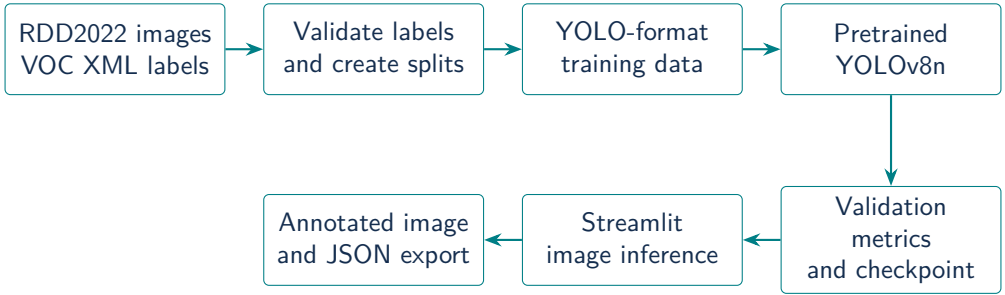
Problem Statement & Objectives

Problem statement

Develop an image-based system that localizes and classifies four road damage types, and evaluates detection quality using a reproducible workflow.

1. Prepare validated annotations and separate train, validation and test sets.
2. Train and evaluate a pretrained YOLOv8n baseline.
3. Implement and later compare paper-inspired architectural variants.
4. Provide an interface for image upload, detection and result export.

Proposed Method: End-to-End Workflow



Implementation: Python, PyTorch, Ultralytics and Streamlit.

Research extension: train individual modifications and the combined PD-inspired model under the same protocol.

Source: Local application, training code and architecture documentation [4].

Proposed Method: Architecture and Implementation Status

Variant	Local implementation	Evidence available
Baseline	Standard YOLOv8n; four-class detection output.	Three real-data epochs.
BOT / LSKA	Residual attention and factorized spatial kernels.	Implemented; smoke checks.
Ghost	Replace four neck C2f blocks with C2fGhost.	Implemented; smoke checks.
LSCD	Shared GroupNorm head across three scales.	Implemented; smoke checks.
PD-inspired	Combine the four modifications.	No local real-data result.

Boundary: these variants are paper-inspired implementations, not an exact reproduction. Their accuracy and efficiency remain to be measured.

Source: docs/architecture.md and local experiment metadata [4].

Proposed Method: Dataset Preparation

6,650 retained India images

- Convert VOC XML boxes to normalized YOLO coordinates.
- Check labels, image dimensions and bounding boxes.
- Skip 1,056 images containing unsupported classes, including D44.
- Check exact image duplicates before splitting.

Split	Images	Share
Train	5,320	80%
Validation	665	10%
Test	665	10%

Seed: 42. No exact duplicates found.

Sequence grouping was not used; nearby frames may still be correlated.

Source: Local split manifest and dataset audit [4]. Test set reserved; no held-out test score reported.

Experimental Setup: Two Different Protocols

	Reference paper [1]	**Local baseline [4]**
Data	10,000 selected images; six countries	6,650 retained India images
Train / validation	8,000 / 2,000	5,320 / 665; 665 reserved for test
Epochs / batch	300 / 16	3 / 4
Input size	640×640	640×640
Optimizer / initial LR	SGD / 0.01	SGD / 0.01
Compute	NVIDIA RTX 2060, 6 GB	CPU
Model	YOLOv8n and YOLOv8-PD	Pretrained YOLOv8n baseline

Interpretation: the two results are contextual references, not a controlled head-to-head comparison.

Evaluation Metrics

Detection quality

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

Precision measures how many predictions are correct. Recall measures how many annotated objects are recovered.

A detection must match both the class and the required box overlap.

Localization and efficiency

IoU: intersection area divided by union area of predicted and reference boxes.

mAP@0.50: mean average precision across classes at IoU 0.50.

mAP@0.50:0.95: average across IoU thresholds 0.50–0.95; stricter localization.

Parameters / GFLOPs: model size and computational cost.

Source: Metric definitions used in the reference paper and Ultralytics evaluation [1,3].

Published Results: YOLOv8-PD

Metric	YOLOv8n	YOLOv8-PD	Change
mAP@0.50	69.2%	70.6%	+1.4 pp
mAP@0.50:0.95	38.0%	39.5%	+1.5 pp
F1 score	0.67	0.68	+0.01
Parameters	3.1 M	2.3 M	≈25.8% fewer
GFLOPs	8.2	6.1	≈25.6% fewer

Published finding: a modest detection improvement with lower model cost. These are the authors' results, not results achieved by this project.

Source: Zeng & Zhong (2024), Tables 1 and 5 [1]. Reductions calculated from rounded table values; pp = percentage points.

Local Results: Preliminary Three-Epoch Baseline

Final-epoch validation

Precision	**37.92%**
Recall	**12.30%**
mAP@0.50	**8.33%**
mAP@0.50:0.95	**3.16%**

Validation trend

Epoch	mAP@.50	mAP@.50:.95
1	2.27%	0.82%
2	5.36%	1.91%
3	8.33%	3.16%

Improvement across three epochs is encouraging, but does not establish convergence or reliable detection.

Source: runs/train/first_model2/results.csv, epochs 1–3 [4]. These are validation log values, not a separate best-checkpoint test evaluation.

Analysis: Imbalance and Limited Training

Objects	Train	Val.	Test
D00	1,118	141	160
D10	50	9	4
D20	1,313	154	168
D40	2,403	279	274

D10 accounts for only about **1%** of training objects.

- Low recall indicates many objects remain undetected.
- Three epochs do not establish convergence.
- Few D10 examples make evaluation unstable.
- Longer training and matched experiments are needed for comparison.

Source: Local audit and validation logs [4]. Causes require further experiments.

Conclusion and Future Work

Completed

- Validated dataset conversion and reproducible split preparation.
- Local baseline training with recorded validation metrics.
- Image upload, detection, statistics and export interface.
- Paper-inspired variants available for further experiments.

Next experiments

- Train to convergence with checkpoint recovery.
- Improve rare-class coverage; assess sequence-aware splits.
- Compare baseline, each module and the combined model with matched settings.
- Evaluate the held-out test set, efficiency and failure cases.

Current conclusion: the prototype works as an experimental pipeline; improved accuracy over the paper has not been demonstrated.

Publications (if any)

No project publication has been reported at this stage.

The selected Scientific Reports article is a reference publication by Jiayi Zeng and Han Zhong; it is not a publication from this project.

Before preparing a research submission

Complete controlled experiments, document implementation differences, evaluate the held-out test set and report reproducible results.

References

1. J. Zeng and H. Zhong, "YOLOv8-PD: an improved road damage detection algorithm based on YOLOv8n model," *Scientific Reports*, vol. 14, article 12052, 2024. doi:10.1038/s41598-024-62933-z.

2. RDD2022 dataset and Global Road Damage Detection Challenge resources. `https://github.com/sekilab/RoadDamageDetector`

3. Ultralytics, YOLOv8 implementation. Local project version: 8.3.203. `https://github.com/ultralytics/ultralytics/tree/v8.3.203`

4. Project evidence: dataset audit, split manifest, architecture notes and local run `first_model2` (experiment metadata, arguments and results CSV).

Source: Institute logo: official NIT Patna website, `https://www.nitp.ac.in/`. Detailed asset and result provenance is included with the source files.